Portfolio 5: distributed and cloud design

The brief permits a discussion or a setup. This section chooses a design discussion, so no cloud deployment is claimed and no cloud account or payment is needed. The same Green Basket Shop database is used throughout.

A simple proposed arrangement

The shop application connects to one primary SQL Server database. That database accepts orders and stock updates. A separately managed secondary copy can support recovery or reporting. A backup is also kept separately. These are proposals; the supplied CREATE DATABASE script builds only the local exercise database.

Replication: keeping another copy

Replication means copying data changes to another database. For example, product and sales data can be copied from the main shop to a reporting server. A reporting copy may be behind the primary, so it should not decide whether the last unit of stock can be sold. A reporting replica alone is not proof of automatic failover.

Sharding: dividing the data

Sharding splits data between databases, for example one shard per branch when the business grows. Each branch would handle its own orders; a central report would combine results. A BranchID and a plan for globally unique order IDs would be needed. This exercise has one shop, so sharding adds unnecessary work and is not implemented.

Availability: staying usable

Availability is the ability to serve users when failures happen. A secondary can help recovery, but it needs monitoring and a tested failover plan. Backups protect against loss and mistakes; they must be restored and tested. A second database on the same machine does not protect against failure of that machine.

Consistency: agreeing on the latest data

All stock-changing requests should go to the primary and use the transaction in PDF 09. A reporting copy can temporarily show an older stock value. Synchronous copying can wait for another server to confirm changes, increasing waiting time. Asynchronous copying can lag, so some recently committed changes may be lost if the primary fails before they reach the recovery copy.

During a network failure

If the primary cannot be reached, the shop should stop confirming new stock deductions rather than let disconnected copies sell the same last item. A cached catalogue may remain viewable, but an order is confirmed only after the primary commits it. This chooses consistent stock decisions over accepting every write during an outage.

Possible cloud deployment

A managed SQL Server-compatible cloud database can reduce routine server maintenance, while a cloud virtual machine gives more control but leaves more administration to the owner. Compare cost, backup retention, recovery needs, access controls and SQL feature support before choosing. A local CREATE DATABASE script does not configure replicas, networking or failover.

Discussion evidence to include

Design decision	Reason for this shop
One primary for writes	Keep sales and stock changes consistent.
Secondary for recovery/reporting	Reduce interruption and separate reporting work.
No sharding yet	One small shop does not justify the extra complexity.
Separate backups and restore tests	Copying changes can also copy mistakes.
Stop stock writes if primary is unavailable	Avoid selling unavailable stock from stale copies.

Submit this discussion as Portfolio 5 evidence. If you later deploy it, add a diagram of the actual topology, redacted configuration screenshots and a genuine recovery or failover test. Never include passwords or connection secrets in GitHub.